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ProteinLM-Augmented Blinded Rediscovery of Food-Derived ACE-Inhibitory Peptides

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Abstract

Food-derived angiotensin-converting enzyme (ACE)-inhibitory peptides are a useful benchmark for computational peptide prioritization because many active sequences have been reported across heterogeneous source materials. We built leakage-controlled blinded candidate universes containing hidden literature positives, matched decoys, random

peptides, and food-protein digestome background. Ranking recipes were frozen before independent holdout curation and evaluated after exact-sequence overlap audits. A v4 frozen broad-shortlist recipe recovered 16 of 31 positives in the top 10 percent of a 56,494-candidate universe. A harder v5 stress panel exposed reduced top-rank recovery. After a post-v5 freeze, the v6 balanced recipe recovered 6 of 52 positives in the top 1 percent of a 58,192-candidate universe, with 11.54-fold enrichment and matched-decoy AUROC 0.7642. ESM2 ProteinLM rescoring improved parent-grouped held-out matched-decoy AUROC from 0.7624 to 0.7832, and a frozen full-universe 0.65 ACE / 0.35 length-calibrated PLM fusion improved matched-decoy AUROC to 0.7813 while preserving 6 of 52 top-1-percent recovery. These results support computational prioritization and rediscovery, not wet-lab potency or receptor-binding claims.

Keywords

bioactive peptides; ACE inhibition; protein language models; ESM2; blinded validation; peptide prioritization; food peptidomics

Declarations

- **Funding:** This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.
- **Competing interests:** The author declares no competing interests.
- **Data availability:** The datasets and analysis code used for this study are available from the corresponding author on reasonable request. Non-sensitive derived summary tables, reports, and reproducibility manifests are included in the submission package where permitted.
- **Ethics:** Not applicable; this is a computational study based on literature-derived peptide evidence and generated analysis artifacts.
- **AI-assisted tools:** Language-model tools were used to assist code implementation, artifact generation, and manuscript drafting. The author remains responsible for study design, curation, code, analysis, interpretation, and final manuscript.

Submission Blockers

- Corresponding author email.
- ORCID if required by the submission system.
- Final decision on whether to replace request-only code/data access with an archival repository link.